

1. If you have a laptop with a three-hole plug to charge, you currently only have a two-hole socket. Now suppose that you have a Laptop class that implements the ThreeHoleIf interface and a PowerSocket class that implements the TwoHoleIf interface. Design a ChargeAdapter class that using the adapter pattern to make the laptop can be charged. Write a Test class to test the result. (Tips: you can refer to the example on pages 127-129)

ChargeAdapter

```
package experiment8_1;

class ChargeAdapter implements ThreeHoleIf{

    private TwoHoleIf twoHoleIf;

    public ChargeAdapter(TwoHoleIf twoHoleIf){
        this.twoHoleIf = twoHoleIf;
    }

    @Override
    public void headWithThreeHoles() {
        twoHoleIf.headWithTwoHoles();
        System.out.println("Adapting two-hole charger to
three-hole charger");
        System.out.println("Laptop is now charging.");
    }
}
```

Laptop

```
package experiment8_1;

class Laptop implements ThreeHoleIf{

    @Override

    public void headWithThreeHoles() {

        System.out.println("The laptop is using three-hole
charger");

    }

}
```

PowerSocket

```
package experiment8_1;

class PowerSocket implements TwoHoleIf{

    @Override

    public void headWithTwoHoles() {

        System.out.println("The power socket is using
two-hole charger");

    }

}
```

Test

```
package experiment8_1;
```

```
public class Test {  
    public static void main(String[] args) {  
        TwoHoleIf powerSocket = new PowerSocket();  
        ThreeHoleIf laptopAdapter = new  
ChargeAdapter(powerSocket);  
        laptopAdapter.headWithThreeHoles();  
    }  
}
```

ThreeHoleIf

```
package experiment8_1;  
  
interface ThreeHoleIf {  
    void headWithThreeHoles();  
}
```

TwoHoleIf

```
package experiment8_1;  
  
interface TwoHoleIf {  
    void headWithTwoHoles();  
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ IDEA
The power socket is using two-hole charger
Adapting two-hole charger to three-hole charger
Laptop is now charging.
```

```
Process finished with exit code 0
```